

Assignment 12

Design and configure a small network topology using GNS3

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Problem Statement:

- Design and configure a small network topology with 2 Switches and a single Router using GNS3.
- Implement basic IP addressing and subnetting.

1. Introduction

This report documents the design and configuration of a small network topology using GNS3 (Graphical Network Simulator 3). The topology consists of two Layer 2 switches and a single Layer 3 router, connected to form two separate subnets. Basic IP addressing and subnetting were applied to enable inter-subnet communication through the router.

GNS3 is an open-source network simulation platform that allows engineers and students to emulate real network hardware and topologies. It supports Cisco IOS images and virtualised devices, making it suitable for hands-on networking experiments without physical equipment.

2. Network Topology Design

2.1 Topology Overview

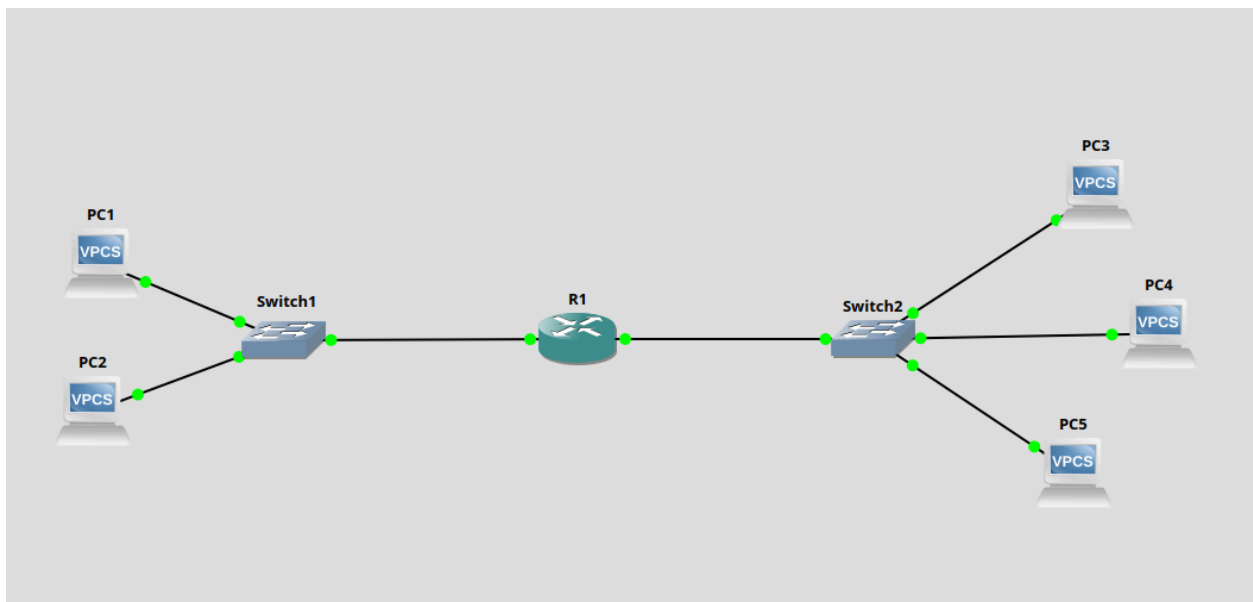
The network consists of the following devices:

- One router (R1) acting as the default gateway for both subnets.
- Two switches (SW1 and SW2), each connecting a group of end hosts to the router.
- Five VPCS hosts: PC1 and PC2 on Subnet A, PC3, PC4, and PC5 on Subnet B.

Switch1 is connected to interface FastEthernet0/0 of R1 and serves subnet 192.168.1.0/24. Switch2 is connected to interface FastEthernet1/0 of R1 and serves subnet 192.168.2.0/24. The router provides inter-subnet routing between the two networks.

2.2 Logical Topology

The logical layout of the topology is as follows:



R1 connects the two broadcast domains via its FastEthernet0/0 and FastEthernet1/0 interfaces, forwarding packets between Subnet A and Subnet B based on its routing table.

3. IP Addressing and Subnetting

3.1 Addressing Scheme

Two /24 subnets from the 192.168.0.0/16 private address range were used. Subnet A (192.168.1.0/24) serves Switch1 with PC1 and PC2. Subnet B (192.168.2.0/24) serves Switch2 with PC3, PC4, and PC5. Each subnet provides 254 usable host addresses, more than sufficient for this lab.

3.2 IP Address Table

Device	Interface	IP Address	Subnet Mask	Network
Router R1	FastEthernet0/0	192.168.1.1	255.255.255.0 (/24)	192.168.1.0/24
Router R1	FastEthernet1/0	192.168.2.1	255.255.255.0 (/24)	192.168.2.0/24
Switch1	—	—	—	192.168.1.0/24
Switch2	—	—	—	192.168.2.0/24
PC1	eth0	192.168.1.2	255.255.255.0 (/24)	192.168.1.0/24
PC2	eth0	192.168.1.3	255.255.255.0 (/24)	192.168.1.0/24
PC3	eth0	192.168.2.2	255.255.255.0 (/24)	192.168.2.0/24
PC4	eth0	192.168.2.3	255.255.255.0 (/24)	192.168.2.0/24
PC5	eth0	192.168.2.4	255.255.255.0 (/24)	192.168.2.0/24

3.3 Subnet Summary

Subnet	Network Address	Broadcast	Usable Range	Hosts
Subnet A (SW1 side)	192.168.1.0	192.168.1.255	.1 – .254	254
Subnet B (SW2 side)	192.168.2.0	192.168.2.255	.1 – .254	254

3.4 Subnetting Rationale

A /24 prefix length (255.255.255.0) was chosen for both subnets because:

- It is the most commonly taught subnet mask and maps cleanly to the third octet.
- It provides 254 usable host addresses, sufficient for small lab networks.
- It keeps the two subnets logically separated with no address overlap.

The router's interfaces serve as the default gateway for each subnet — 192.168.1.1 for Subnet A and 192.168.2.1 for Subnet B.

4. Device Configuration

4.1 Router R1 Configuration

The following IOS commands were used to configure the router interfaces:

```
R1# configure terminal
R1(config)# interface FastEthernet0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface FastEthernet1/0
R1(config-if)# ip address 192.168.2.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# end
```

4.2 Switch Configuration

The switches (SW1 and SW2) were configured with management IP addresses on VLAN 1 for administrative access, and their default gateways were set to the router's corresponding interface.

```
SW1# configure terminal
SW1(config)# interface vlan 1
SW1(config-if)# ip address 192.168.1.2 255.255.255.0
SW1(config-if)# no shutdown
SW1(config-if)# exit
SW1(config)# ip default-gateway 192.168.1.1
```

SW2 was configured identically using the 192.168.2.0/24 addressing scheme, with 192.168.2.1 as its default gateway.

4.3 Host Configuration

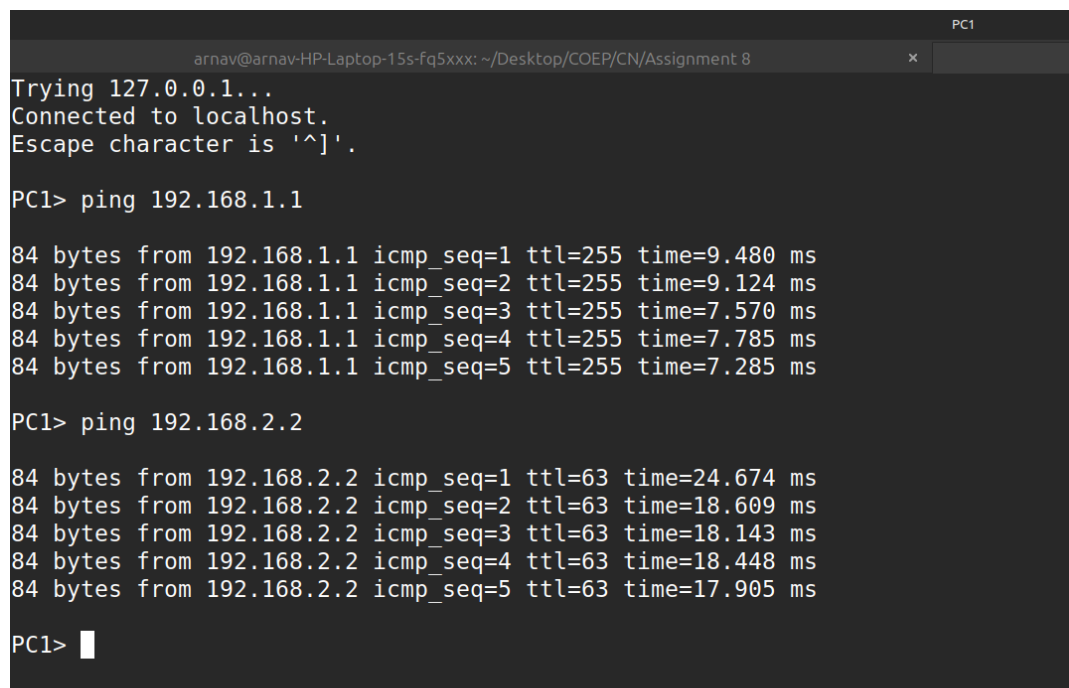
The switches (SW1 and SW2) were configured with management IP addresses on VLAN 1 for administrative access, and their default gateways were set to the router's corresponding interface.

5. Connectivity Verification

5.1 Ping Tests

Connectivity was verified using the ping command from each PC. The following tests were performed:

- PC1 → R1 f0/0 (192.168.1.1): Success — confirms local subnet connectivity from Switch1 side.
- PC3 → R1 f1/0 (192.168.2.1): Success — confirms local subnet connectivity from Switch2 side.
- PC1 → PC3 (192.168.2.2): Success — confirms inter-subnet routing through R1.
- PC3 → PC1 (192.168.1.2): Success — confirms bidirectional routing.
- PC2 → PC5 (192.168.2.4): Success — confirms routing works across all hosts on both subnets.



```
arnav@arnav-HP-Laptop-15s-fq5xxx: ~/Desktop/COEP/CN/Assignment 8
Trying 127.0.0.1...
Connected to localhost.
Escape character is '^]'.

PC1> ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=255 time=9.480 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=255 time=9.124 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=255 time=7.570 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=255 time=7.785 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=255 time=7.285 ms

PC1> ping 192.168.2.2

84 bytes from 192.168.2.2 icmp_seq=1 ttl=63 time=24.674 ms
84 bytes from 192.168.2.2 icmp_seq=2 ttl=63 time=18.609 ms
84 bytes from 192.168.2.2 icmp_seq=3 ttl=63 time=18.143 ms
84 bytes from 192.168.2.2 icmp_seq=4 ttl=63 time=18.448 ms
84 bytes from 192.168.2.2 icmp_seq=5 ttl=63 time=17.905 ms

PC1> █
```

```

arnav@arnav-HP-Laptop-15s-fq5xxx: ~/Desktop/COEP/CN/Assignment 8
Trying 127.0.0.1...
Connected to localhost.
Escape character is '^]'.

PC3> ping 192.168.2.1

84 bytes from 192.168.2.1 icmp_seq=1 ttl=255 time=3.631 ms
84 bytes from 192.168.2.1 icmp_seq=2 ttl=255 time=8.608 ms
84 bytes from 192.168.2.1 icmp_seq=3 ttl=255 time=7.463 ms
84 bytes from 192.168.2.1 icmp_seq=4 ttl=255 time=7.099 ms
84 bytes from 192.168.2.1 icmp_seq=5 ttl=255 time=8.503 ms

PC3> ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=63 time=25.832 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=63 time=18.807 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=63 time=17.471 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=63 time=18.698 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=63 time=18.518 ms

PC3> █

```

5.2 Routing Table

The show ip route command on R1 confirmed that both directly connected subnets were present in the routing table:

```

R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.1.0/24 is directly connected, FastEthernet0/0
C    192.168.2.0/24 is directly connected, FastEthernet1/0
R1# █

```

Both routes are marked C (Connected), indicating no dynamic routing protocol was needed for this topology.

6. Switch vs. Router Comparison

The following table compares the roles of the switches and router used in this topology:

Feature	Switch	Router
OSI Layer	Layer 2 (Data Link)	Layer 3 (Network)
Forwarding Basis	MAC Address Table	IP Routing Table
Broadcast Domain	Single domain per VLAN	Separates broadcast domains
Inter-subnet Routing	Not supported	Supported (gateway role)
Used In This Lab	SW1 and SW2	R1

In this lab, the switches operated entirely at Layer 2, forwarding frames within each subnet based on MAC addresses. The router performed Layer 3 forwarding, consulting its IP routing table to move packets between the two subnets.

7. Observations

- GNS3 accurately emulated the behaviour of Cisco IOS switches and routers, producing results consistent with physical hardware.
- Without the correct default gateway set on each VPCS host, inter-subnet pings failed, confirming that Layer 3 routing is required for cross-subnet communication.
- The GNS3 ethernet_switch nodes required no configuration. They forwarded traffic within each subnet automatically, demonstrating plug-and-play Layer 2 operation.
- The no shutdown command was essential on R1's FastEthernet0/0 and FastEthernet1/0 interfaces; Cisco IOS interfaces default to administratively down.
- The show ip route output confirmed that directly connected routes are populated automatically when interfaces are configured and brought up.

8. Conclusion

- This assignment demonstrated the design and configuration of a small two-subnet network using GNS3 with two switches and a single router. IP addressing was applied using the 192.168.0.0/16 private range with /24 subnets, providing clear logical separation between the two network segments.
- The router served as the inter-subnet gateway, enabling PC1 and PC2 to communicate across different subnets. Ping tests confirmed full bidirectional connectivity, and the routing table verified that both subnets were reachable as directly connected routes.
- The experiment reinforced key networking concepts: the role of default gateways, the difference between Layer 2 switching and Layer 3 routing, and the practical steps involved in bringing up interfaces in Cisco IOS. GNS3 proved to be an effective tool for simulating these scenarios without requiring physical hardware.